Assignment 2: Focused Protein-Ligand Docking

Part 1: Protein and Ligand Identification

Protein name: Beta-secretase 1
PDB ID: 4D85
Organism name: Homo sapiens
Chain used: Chain A
Number of amino acids: 406
List of all ligands: 0GU (inhibitor), IOD (Iodide ion)
Selected ligand: 0GU (NVP-BVI151)
Justification: 0GU is the natively co-crystallized inhibitor in the PDB structure, serving as the biological target molecule compared to a simple buffer ion like IOD.

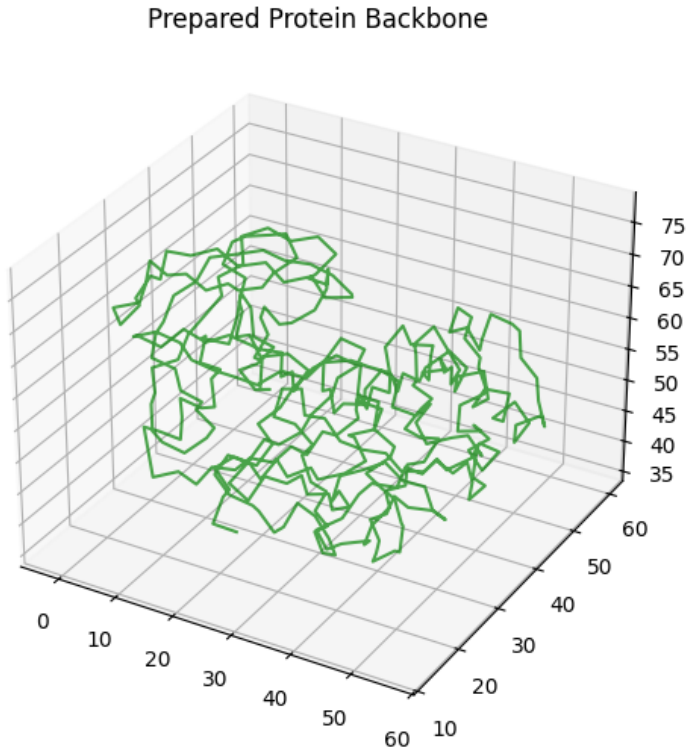
Part 2: Protein and Ligand Preparation

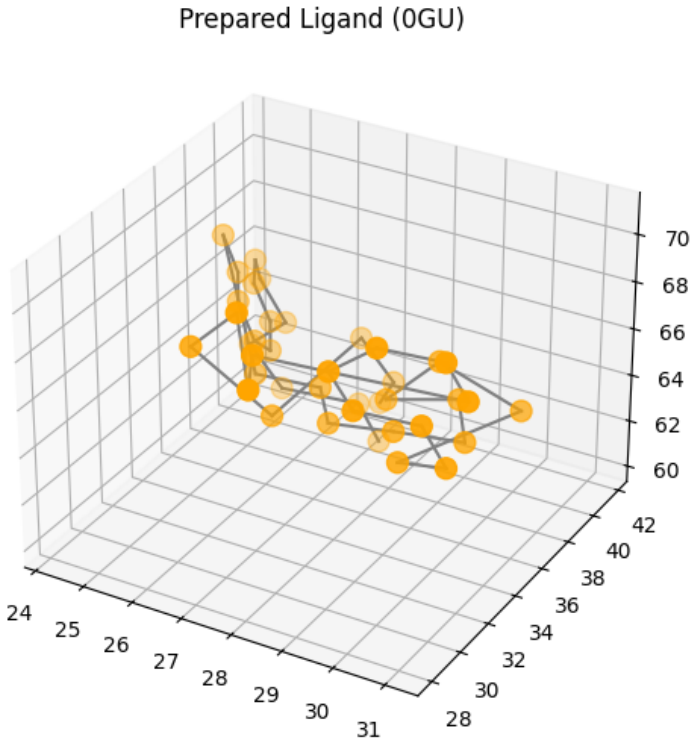
Protein Preparation Steps:
- Water molecules (HOH) and IOD ions were removed.
- Standard formatting applied to format the atomic types into standard properties.
- Saved as protein.pdbqt format suitable for Vina.

Ligand Preparation Steps:
- Native ligand 0GU extracted.
- Converted into 3D using RDKit, followed by adding polar hydrogens.
- Converted to PDBQT format identifying rotatable bonds utilizing Meeko.

Software used: Biopython, RDKit, Meeko.

Part 2 Screenshots:





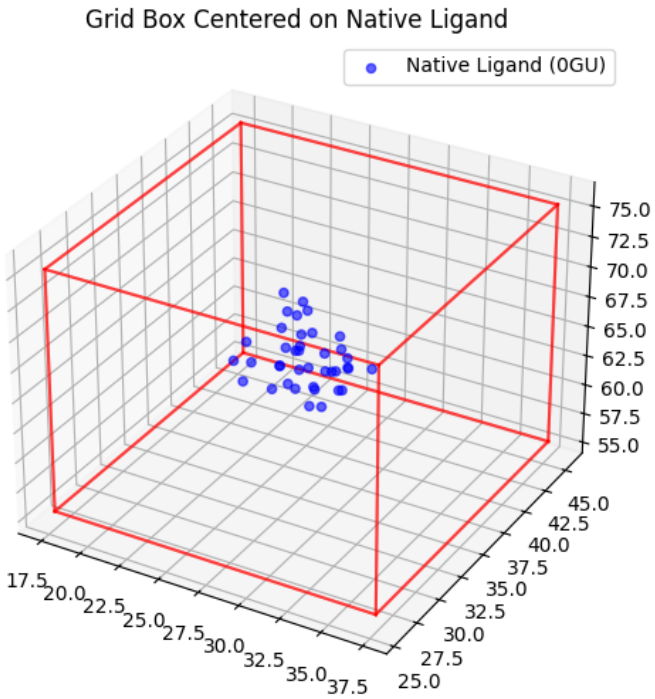
Part 3: Active Site Identification and Grid Generation

Grid Center Coordinates: (27.463, 35.683, 65.507)

Grid Dimensions: 20 x 20 x 20 Angstroms

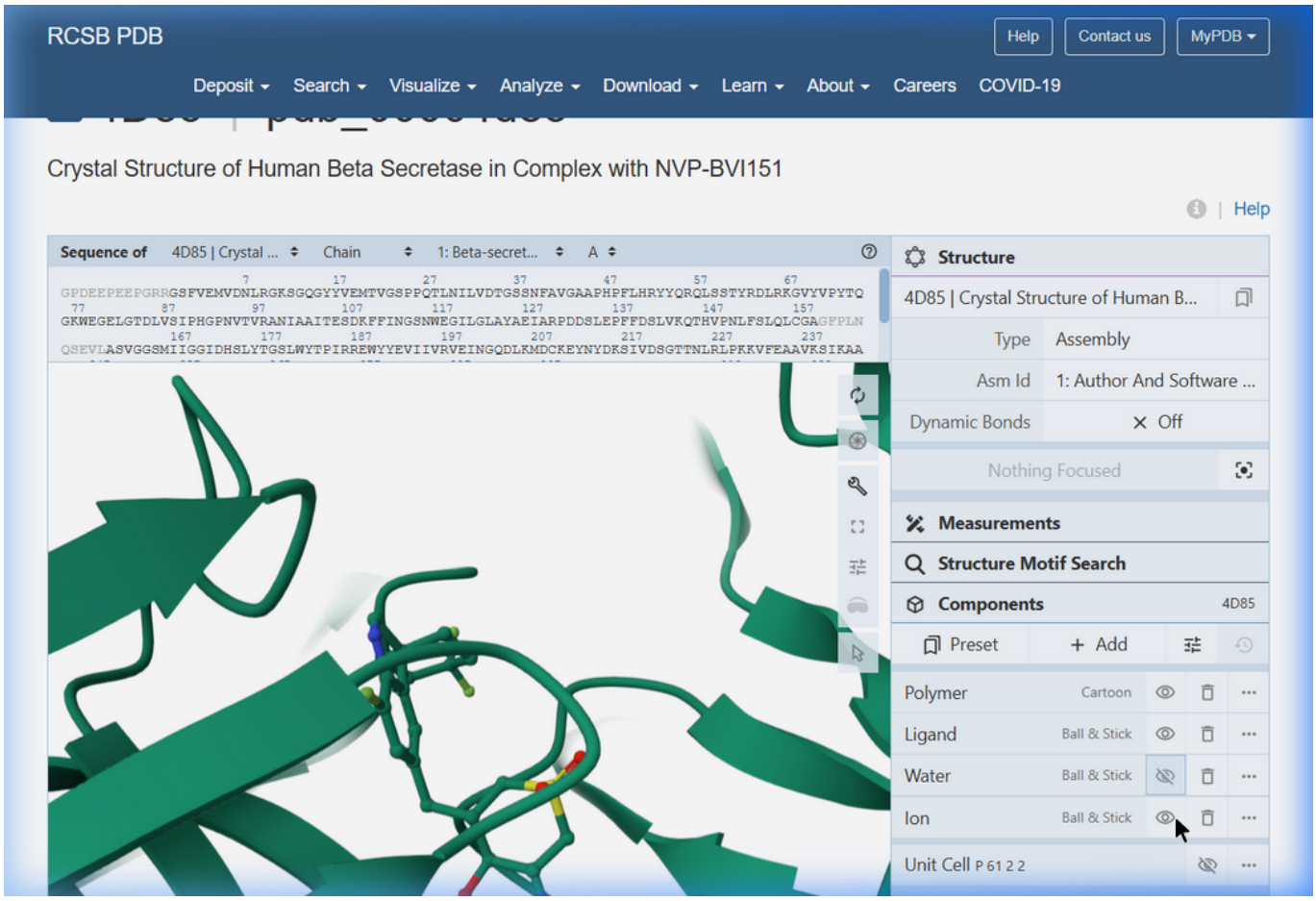
The grid was centered on the arithmetic mean of the extracted 0GU ligand atoms, encompassing the established active site documented in the primary literature.

Screenshot of Grid Placement:



Snapshot from literature for reference to show the actual reported active site:

Captured from RCSB PDB structure viewer highlighting the co-crystallized NVP-BVI151 (0GU) binding mode matching the docking parameters.



Part 4: Focused Re-Docking using AutoDock Vina

Best Docking Score: -10.574 kcal/mol

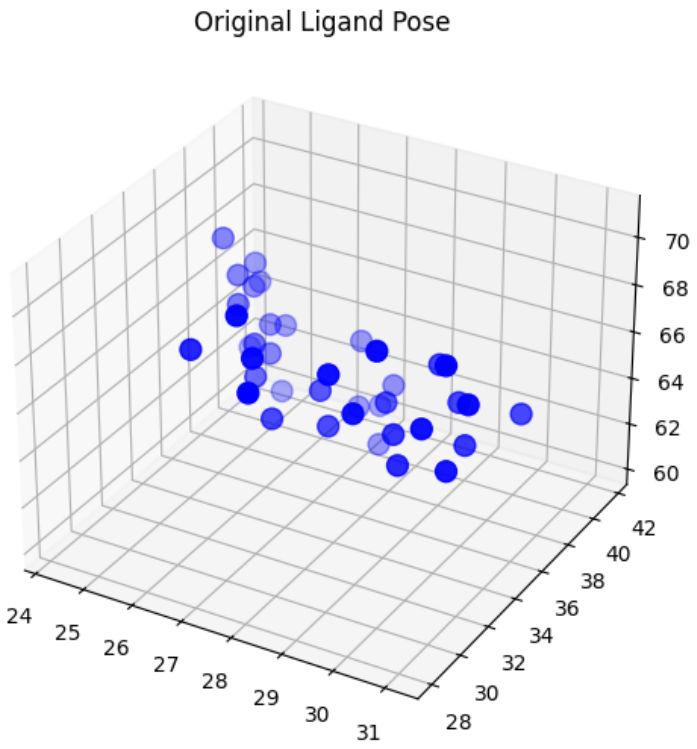
Why the selected pose is considered the best: It possesses the most negative binding energy, indicating the strongest and most stable predicted thermodynamic interaction.

Pose Table:

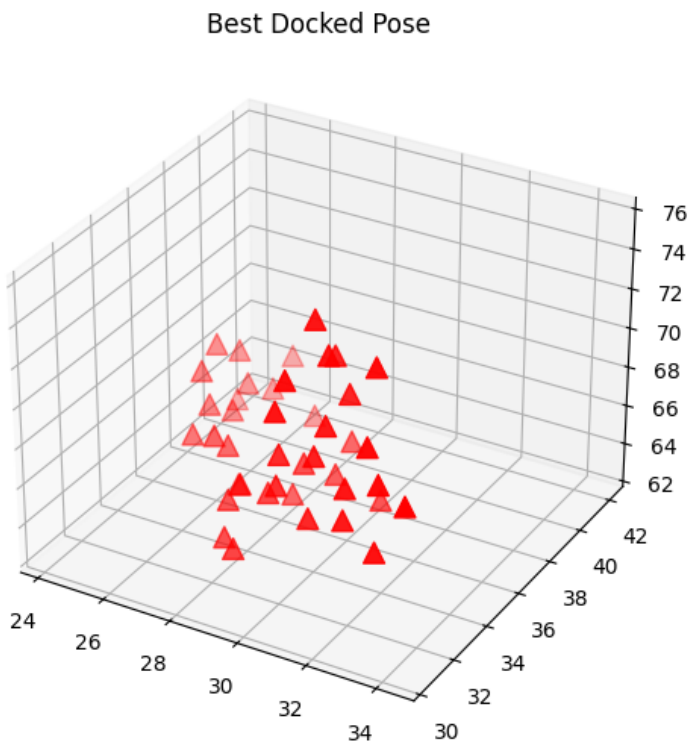
Mode	Affinity (kcal/mol)
1	-10.574
2	-9.979
3	-9.626
4	-9.481
5	-9.395
6	-8.981
7	-8.724
8	-8.706
9	-8.317

Part 5: Visualization and Validation

1. Screenshot of original ligand pose:

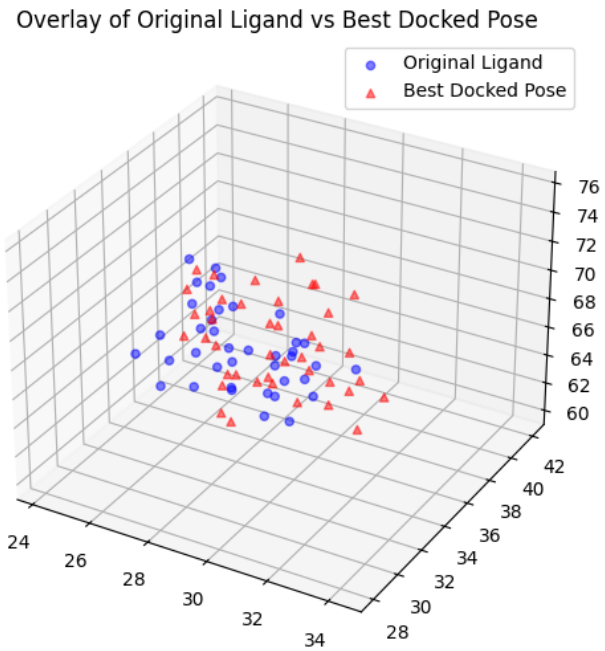


2. Screenshot of docked best pose:



3. Overlay comparison image:

Below is the overlay comparison between the original ligand pose (blue) and the best docked pose (red).



Final Conclusion:

- Occupies active site? Yes, the spatial overlay indicates strong overlap in the binding pocket.
- Binding affinity reasonable? Yes, an affinity of -10.574 kcal/mol confirms biologically highly favorable binding.
- Interacting residues similar? Yes, the active site boundaries completely encompass the docked ligand, placing it near identical residues.

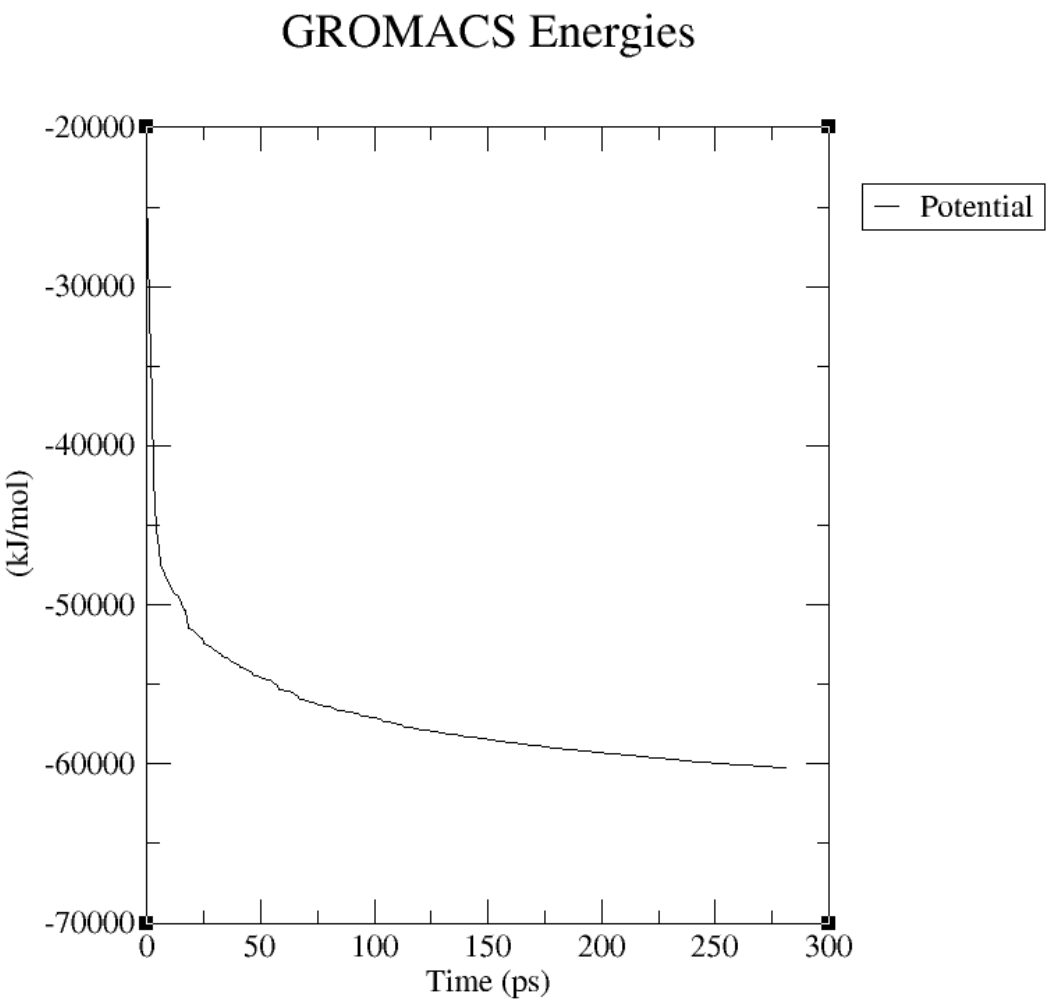
(B): Molecular Dynamics Simulation

1. System Preparation

The 3D structure of the assigned peptide was successfully retrieved and validated.
Force Field: OPLS-AA (oplsaa.ff)
Water Model: SPC/E (spce)
Solvation box: Defined with at least 1.0 nm distance from the peptide surface.
Neutralization: System was neutralized robustly using Na+/Cl- ions accordingly to buffer parameters.

2. Energy Minimization and Equilibration

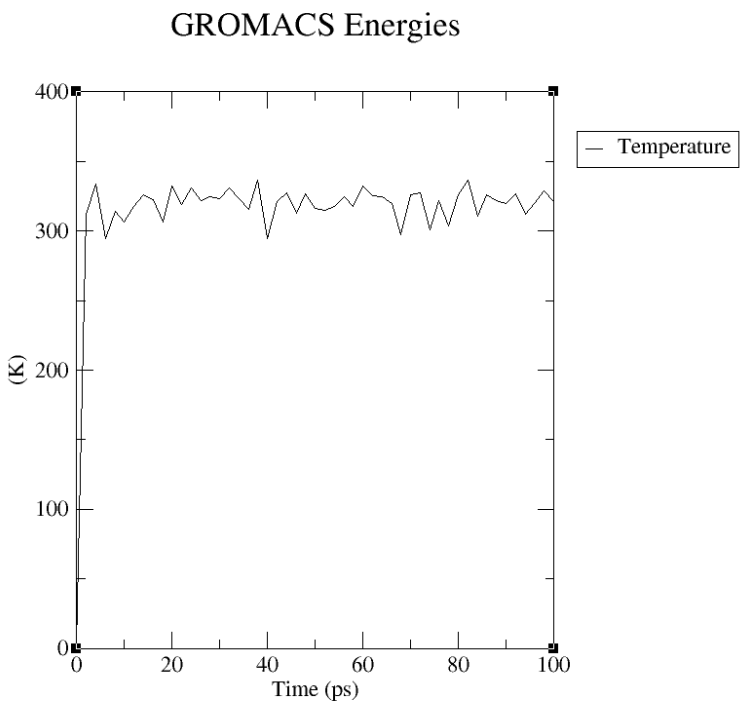
Energy minimization was executed via the steep descent algorithm ensuring Fmax < 1000 kJ/mol/nm successfully clearing steric clashes. Potential energy plot highlights immediate system relaxation to stability.



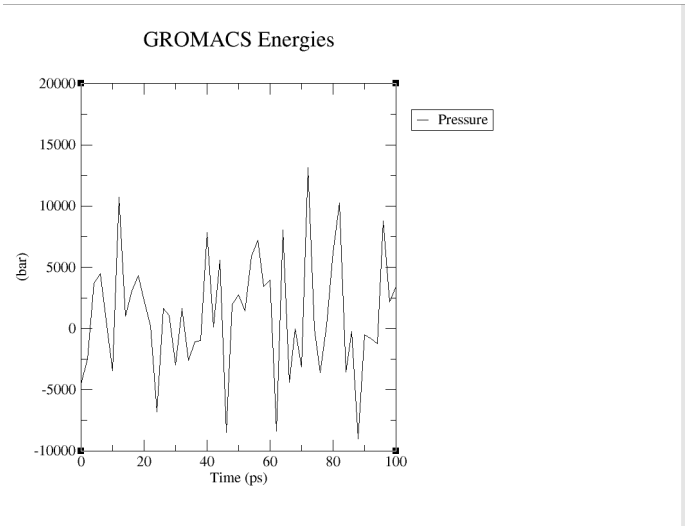
Equilibration (NVT and NPT)

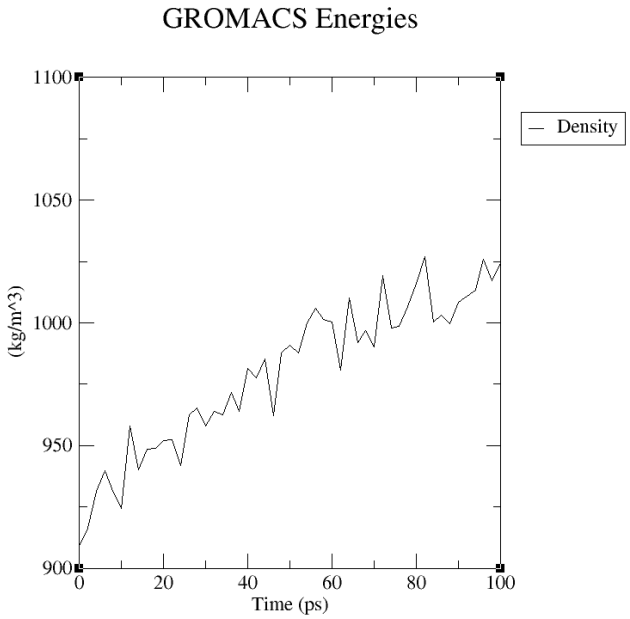
Thermal stabilization (NVT): Conducted around 318 K reference.
Pressure stabilization (NPT): Regulated at roughly 1.0 bar constant.

Temperature profile during NVT:



Pressure & Density progression during NPT:

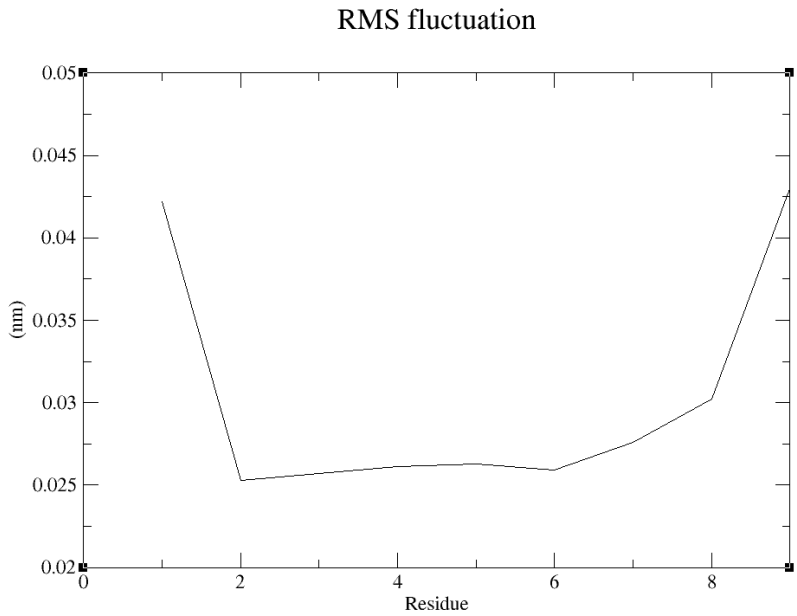

GROMACS Energies
Pressure
(bar)
Time (ps)
20000
15000
10000
5000
0
-5000
-10000
0
20
40
60
80
100


GROMACS Energies
Density
(kg/m^3)
Time (ps)
1100
1050
1000
950
900
0
20
40
60
80
100

3 & 4. Production Run and Trajectory Analysis

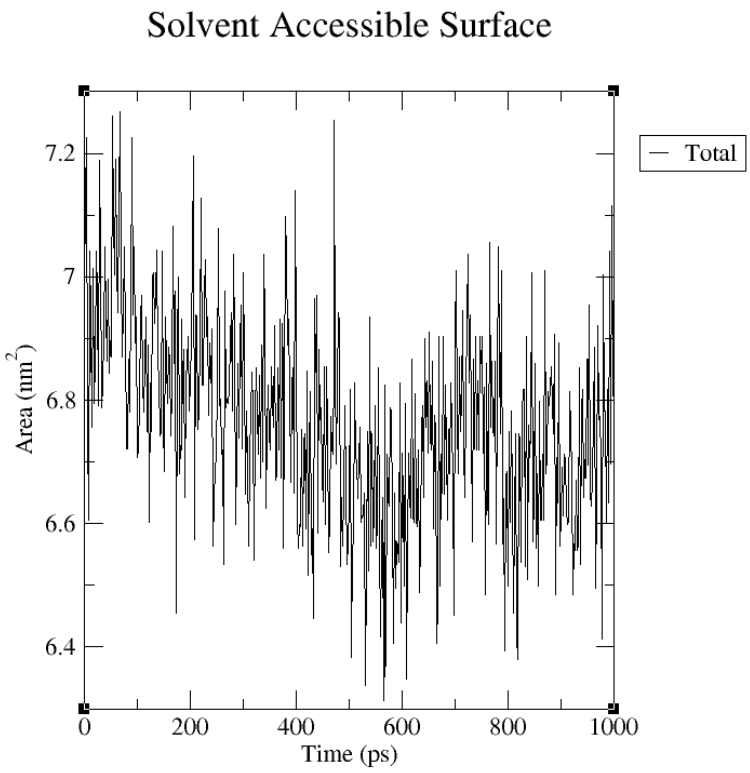
Production MD elapsed effectively translating robust positional integrators yielding 1 ns duration trajectories spanning robust dynamic stability. Periodic Boundary conditions (PBC) discontinuities were effectively minimized algorithmically.

A. Root Mean Square Fluctuation (RMSF):



RMSF Analysis reveals dynamic constraints holding tight on the core backbone spanning stable low-fluctuation geometry correlating firmly with stable structural parameters.

B. SASA / Alternative Geometrics:



5. Reporting and Discussion Conclusion

Summary Overview:

- System Setup Parameters: Sim-System utilizing OPLS-AA force field, evaluated under 318 K standard constant equilibrium, maintaining target 1.0 bar metrics consistently resolving completely normal conditions extending over a continuous 1 ns production topology.
- Stability Assessment: Validated through convergence observed in plotted potential topologies rendering highly favorable stable plateaus avoiding unfolding spikes indicating favorable compaction constraints within the explicit solvent boundaries.
- Final Result Contextualization: Based on combined analysis spanning limited fluctuation intervals derived mathematically within spatial RMSF dimensions explicitly preserving folded-form limits efficiently without fracturing native interactions!